

FACTFULNESS

Ten Reasons We're Wrong About the World
—and Why Things are Better Than You Think

HANS ROSLING



The Big “So What”

Most people—including business/political leaders and professionals—have grossly inaccurate views of the world we live in, and see the world to be much worse than it really is. **This book helps us to recognize and manage 10 mental filters that distort our perceptions, and to develop a fact-based worldview so we can avoid unnecessary stress and improve our ability to make sound decisions.**



Introduction



Over the years, Rosling presented various multiple-choice questions about global trends (e.g. poverty, health, environment) to thousands of people worldwide. Even if people randomly chose 1 of the 3 answers provided, they should've gotten 33% of the questions right. Yet, consistently less than 33% of the questions were answered correctly, i.e. we are systematically wrong in our worldview.

- For example, in the last 20 years, the proportion of the world population living in extreme poverty has almost halved. Yet, only 7% of the people surveyed got that right; the majority believed that the proportion of people living in extreme poverty has remained the same, or has almost doubled.
- Generally, people believe the world is getting worse, when facts and data show that the world is getting better. More girls in low income countries are going to school, children are getting vaccinated, most people have moved out of extreme poverty, and child mortality rates have decreased in every country worldwide.



In reality, most of us don't see the world as it is, because we sift the inputs through a mental filter that favors dramatic information.

KEY QUOTES

“Every group of people I ask thinks the world is more frightening, more violent, and more hopeless—in short, more dramatic—than it really is.”

“Step-by-step, year-by-year, the world is improving... Though the world faces huge challenges, we have made tremendous progress. This is the fact-based worldview.”

- Thus, we tend to have an exaggerated and overly-negative view of the world, which creates unnecessary fear and stress. We also can't make sound decisions if we're operating on wrong facts.
- In this summary, we'll explain the 10 “dramatic instincts” that all humans have, how to control these instincts and apply factfulness to see the world more accurately.

The 10 Dramatic Instincts

1. The Gap Instinct

We tend to divide things into 2 distinct groups and imagine a gap between them, e.g. good vs bad, rich vs poor. This is one of most damaging instincts as it creates an imagined chasm that leads to all sorts of misperceptions.

Most of us still mentally divide the world into 2 parts: developing non-Western countries vs developed Western countries. This notion is at least 20-30 years outdated.

- We have a wrong mental picture of how people in poor non-Western countries live (e.g. life expectancy, water quality, vaccination rates, under-nourishment); and
- We over-estimate how much of the world population live in these low-income countries.

In short, we imagine a big gap between the developed and developing countries, with most people living in dire conditions. In reality, most people are living in the middle where we *think* the gap is.

It's more meaningful to look at the world with 4 levels of income:

World Population & Income levels as of 2017

LEVEL 1	LEVEL 2	LEVEL 3	LEVEL 4
\$2	\$8	\$32	
• Drinking water: Mud-holes • Transport: Walk barefoot • Cooking: Using firewood • Food: Eat what you grow/find • Other: High child mortality rates (can't afford antibiotics)	• Drinking water: Buckets • Transport: Sandals, bicycles • Cooking: Small gas stove • Food: Buy some food/rear chicken • Other: Unstable electricity, light bulbs	• Drinking water: Outdoor tap • Transport: Motorcycles • Cooking: Proper stove • Food: Fridge, variety of dishes • Other: Can afford medical bills & high school education for kids	• Drinking water: Hot/cold water indoors • Transport: Cars, airplanes • Cooking: Full kitchen • Food: Huge variety • Other: >12 years of education

= 1 bil people

Income: per person per day, adjusted for price differences

Source: Gapminder

KEY QUOTES

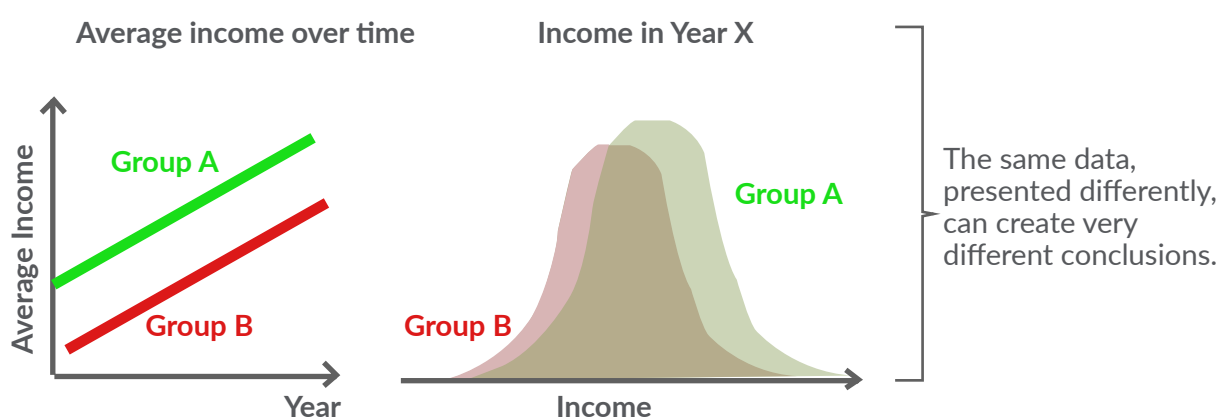
“The idea of a divided world with a majority stuck in misery and deprivation is an illusion. A complete misconception. Simply wrong.”

- People living in extreme poverty (Level 1) suffer from high child mortality rates, malnutrition and terrible living conditions.
- However, once they can earn \$2 a day, they'll cross over to the next level and can afford some basic essentials. By the time they get to Level 3, they're usually working multiple jobs and have decent savings to enjoy choices in life and send their kids to high school. The majority (5 out of 7 people) today are living on Levels 2 and 3.
- If you're reading this summary, you're probably living on Level 4, with at least 12 years of education, an income of at least \$32 a day and access to a range of modern amenities.
- Everyone in human history started at Level 1 and it usually takes generations for a family to move from Level 1 to Level 4. About 200 years ago, 85% of the population was living on Level 1. While it's bad that some people are still living in extreme poverty today, the overall situation is certainly getting better, not worse.

❗ When we split the world into 2 distinct groups with a big gap in between, it suggests conflict. In reality, things usually fall on a spectrum—there're commonalities between groups with no clear lines in between.

📊 There are 3 types of "gap stories" that trigger our gap instinct:

- Comparing averages. Averages help to simplify information, but draw our attention to the gap between 2 sets of numbers. For example, when you compare the average incomes between 2 groups (see the left diagram below), it creates the perception that there's a clear separation between the groups, when in reality the areas of overlap may be greater than the differences (see the right diagram below).



KEY QUOTES

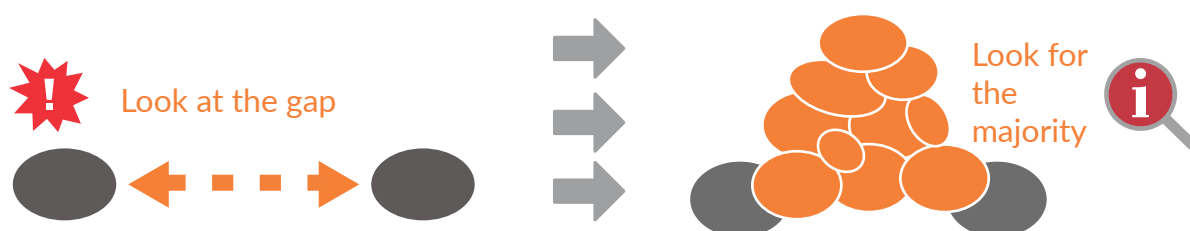
"Dividing the world into two distinct sides is simple and intuitive, and also dramatic because it implies conflict, and we do it without thinking, all the time."

"The gap instinct makes us imagine division where there is just a smooth range, difference where there is convergence, and conflict where there is agreement."

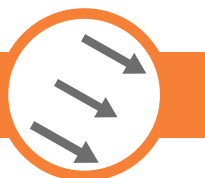
- Comparing extremes. We tend to remember extreme examples (e.g. the richest vs poorest people) as these are more dramatic. However, such extremes are usually the minority and do not reflect the majority which are in the middle.
- View from above. From the top of a tall building, the objects on the ground level look deceptively small. Likewise, when you're living on Level 4, your perception of lives on Levels 1-3 is distorted. Your information on other levels are filtered through the mass media which tends to focus on the extreme and the dramatic.

i To control this instinct, realize that in reality, there's usually no gap, extremes are the minority, and the majority are in the middle where the gap is supposed to be. When you hear a story of 2 distinct groups with a gap between them, examine the spread of data and you're likely to find 2 overlapping groups with no real gap.

CONTROLLING THE GAP INSTINCT



2. The Negativity Instinct



We tend to instinctively notice the bad more than the good.



Global trends. Most people believe that the world is becoming worse, when in reality it's becoming better.

- In the last 20 years, the proportion of people living in extreme poverty has fallen from 29% to 9%, while average life expectancy has increased from 30 to 70 years old. Rosling also presented charts of 16 bad things decreasing globally (e.g. legal slavery, oil spills, HIV infections, ozone depletion) and 16 good things increasing globally (e.g. new movies per year, women's right, electricity coverage, child cancer survival).

KEY QUOTES

"The fact that extremes exist doesn't tell us much. The majority is usually to be found in the middle, and it tells a very different story."

"The reality is often not polarized at all. Usually the majority is right there in the middle, where the gap is supposed to be."

- He showed Sweden's health and wealth statistics from 1800 to 2017, and how they corresponded with other countries' statistics in 2017. In a nutshell, Sweden as a country had moved from Level 1 to Level 4 progressively over the generations, just like other countries. Specifically, 1921 Sweden was comparable to 2017 Zambia, 1947 Sweden was comparable to 2017 Egypt, while 1975 Sweden was comparable to 2017 Malaysia.

 Our overly-negative worldview comes from 3 key factors:

- Selective news & reporting. The mass media tends to cover dramatic stories about wars, disasters, scandals or accidents—rather than gradual improvements—as those stories are more attention-grabbing. Activists and lobbyists also play up short-term dips and play down long-term growth. Thus, we get the impression that things are deteriorating when they're actually improving.
- Poor recall of the past. We avoid reminding ourselves and our children of past miseries and thus remember the past to be better than it really was.
- We feel bad about celebrating improvements when things still seem so bad.

 To control this instinct, remember that you're more likely to receive negative than positive news.

- Expect bad news, but ask yourself if there could be good news of equal or greater scale that haven't reached you. You can also validate the real picture using statistics.
- Realize that things can be both "bad" and "better" at the same time, i.e. the situation has improved from the past, but still needs improvement. With an objective viewpoint, you can take constructive actions rather than lose hope for the future.
- Beware of over-glorified histories and don't hide the past to protect your children.

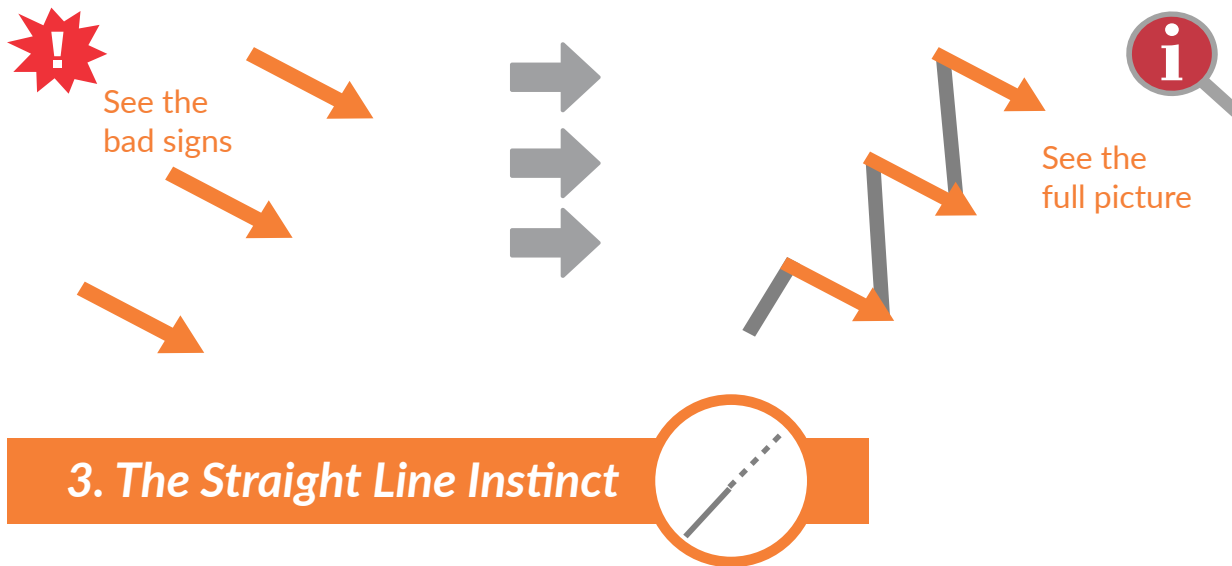
KEY QUOTES

"The positive changes may be more common, but they don't find you. You need to find them."

"Remember: things can be bad, and getting better. Getting better, but still bad."

"The loss of hope is probably the most devastating consequence of the negativity instinct and the ignorance it causes."

CONTROLLING THE NEGATIVITY INSTINCT



3. The Straight Line Instinct

When we see a line going up steadily, we tend to assume the line will continue to go up in the foreseeable future.



Most people think the world population will continue to grow rapidly, with the bulk of the growth coming from babies born in poor, developing countries. In reality, the population will grow because people are now living longer, not because more babies are born. As families move out of extreme poverty, they tend to have fewer babies, since women are educated about contraceptives and no longer need so many kids for manual labor or to replace children who die from diseases or malnutrition. As of 2017, most families have an average of only 2 children.



In reality, most lines will bend at some point.

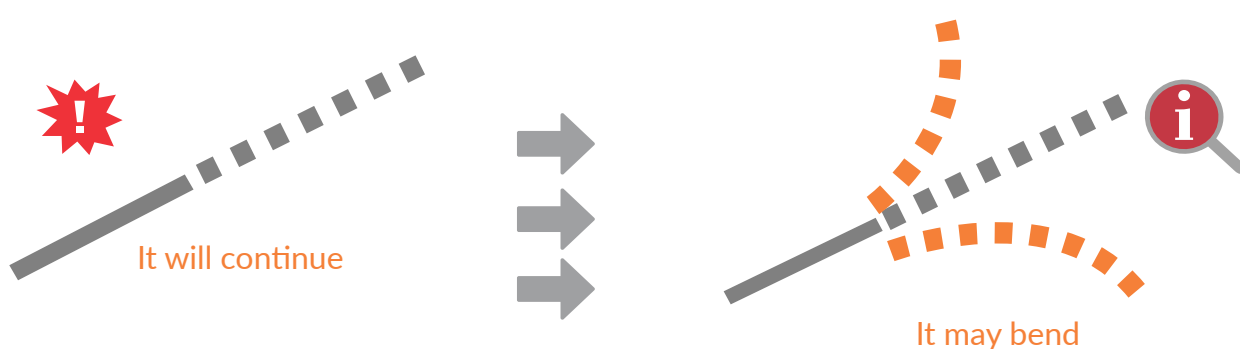
- Certain trends or relationships may follow a straight line, such as the relationship between wealth and health, and the relationship between income and education or recreation spending.
- However, most charts follow other shapes. *S-shaped curves* start out relatively flat, then go up steeply before flattening out again (e.g. the relationship between income and vaccination). *Slides* look like reverse S-curves. *Humps* show trends that get worse before they get better (e.g. the relationship between income and dental health). *Doubling lines* show numbers doubling to exponential growth in the long term (e.g. the relationship between income and distances travelled).

KEY QUOTES

"To understand a phenomenon, we need to make sure we understand the shape of its curve. By assuming we know how a curve continues beyond what we see, we will draw the wrong conclusions and come up with the wrong solutions."

i To control this instinct, remember that most curves and trends come in different shapes, e.g. S-bends, humps or doubling lines. Know the shape or direction of a trend to find the right solution.

CONTROLLING THE STRAIGHT LINE INSTINCT



4. The Fear Instinct



We tend to perceive the world to be scarier than it really is.

! Most of our fears (of violence, contamination, captivity etc.) are exaggerated due to mass media influences and our mental filters.

- The media tends to cover topics like natural disasters, terrorist attacks and diseases because these attract our attention and response. Images of dead and injured people trigger feelings of fear and sorrow, creating an emotional, illogical urge for us to take action (e.g. share the news or donate money).
- Rosling presented detailed statistics to prove that these events are much less threatening than we think. In reality, only 0.1% of all deaths come from natural disasters, 0.001% from plane crashes, 0.7% from murders, 0.05% from terrorism, and 0% from nuclear leaks. Yet, such events get disproportionate coverage on the mass media.

i To control this instinct, realize that there's a difference between what's frightening (i.e. poses a *perceived risk*) and what's dangerous (i.e. poses a *real risk*).

- Something is truly risky only if it's dangerous *and* you're exposed to it. Calculate the actual risk of an event with this formula:

$$\text{Risk} = \text{danger} \times \text{exposure}.$$

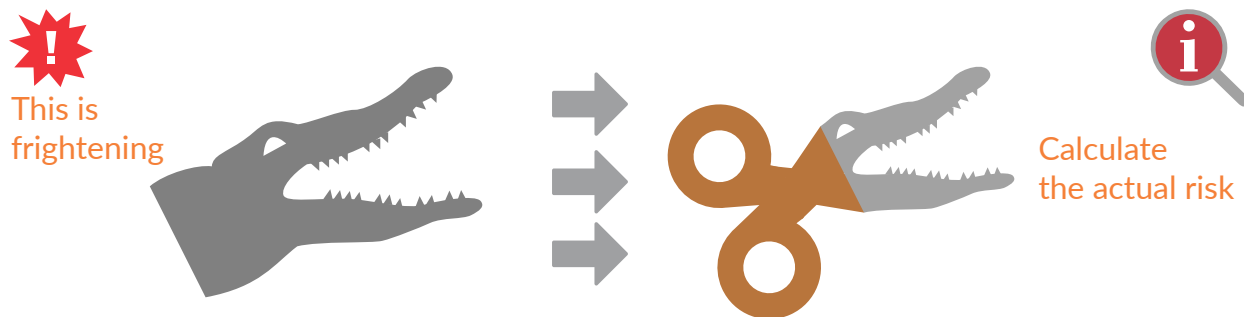
KEY QUOTES

"There's no room for facts when our minds are occupied by fear."

"Here's the paradox: the image of a dangerous world has never been broadcast more effectively than it is now, while the world has never been less violent and more safe."

- Don't let frightening news dictate your actions. Calm yourself down first to resume your analytical abilities, then adopt a fact-based perspective to make informed, effective decisions.

CONTROLLING THE FEAR INSTINCT



5. The Size Instinct



We tend to see things out of proportion, over-estimating (a) the importance of a single event or person that's visible to us, and (b) the scale of an issue based on a standalone number.

- Our resources are limited and it's sub-optimal to focus them on the visible problems in front of you. It's better to step back to see the big picture, then tackle the root of the problem for a bigger impact. For example, a doctor in a Level 2 city can save many more lives if he diverted some of his time and resources to train village health workers and vaccinate children *outside* the hospital, rather than focusing only on the patients *in* the hospital.

i To control this instinct, never interpret a number on its own. Instead, compare and divide numbers to see things in proportion:

- Compare the numbers. In 2016, 4.2mil babies died from preventable diseases; this sounds massive until you compare it with 14.4mil deaths in 1950. Use the 80/20 rule to identify where 80% of your problem cases are, then focus your resources there.
- Divide the numbers. Make a large number more meaningful by dividing it against another number to get a rate. For example the number of babies dying divided by the number of births gives you the child mortality rate. Total carbon

KEY QUOTES

"The fear instinct...makes us give our attention to the unlikely dangers that we are most afraid of, and neglect what is actually most risky."

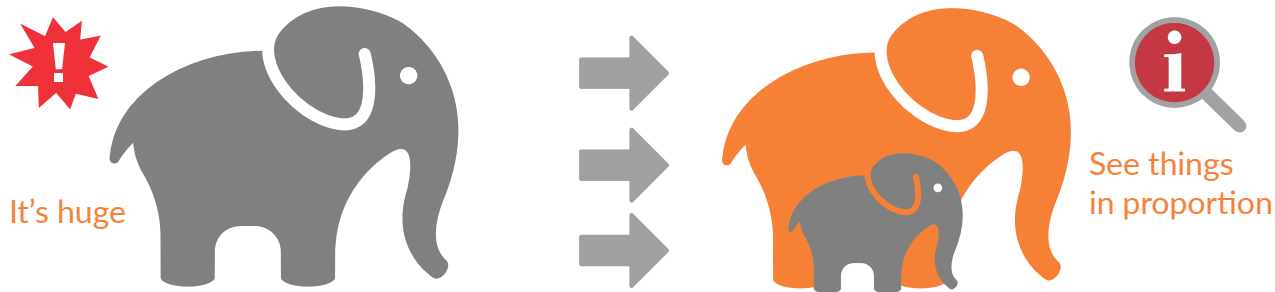
"The size instinct directs our limited attention and resources toward those individual instances or identifiable victims, those concrete things right in front of our eyes."

"To know how to act, and how to prioritize resources, nothing can be more important than doing the cool-headed math and realizing what works and what doesn't."

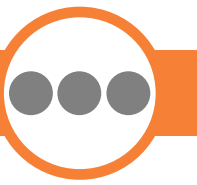
"So long as resources are not infinite—and they never are infinite—it is the most compassionate thing to do to use your brain and work out how to do the most good with what you have."

emission divided a country's population gives you a per-capita rate, which is more meaningful than a large number without context.

CONTROLLING THE SIZE INSTINCT



6. The Generalization Instinct



Categories are needed for us to handle the overwhelming amount of detail around us. However, **we tend to wrongly assume that everything or everyone in a category is similar.**

! We subconsciously put people into “us” vs “them” categories and make all sorts of wrong assumptions about “them”.

- For example, businesses often dismiss Level 2 and 3 markets as “poor countries”, thus missing out on large numbers of consumers with buying power.
- When we make wrong assumptions about others based on our own experiences, it's ineffective and even dangerous. During a visit to a Level 2 hospital, one of Rosling's students tried to stop the lift-doors from closing by sticking her leg between the doors—she almost lost her leg on the assumption that all lift-doors had closing-sensors.

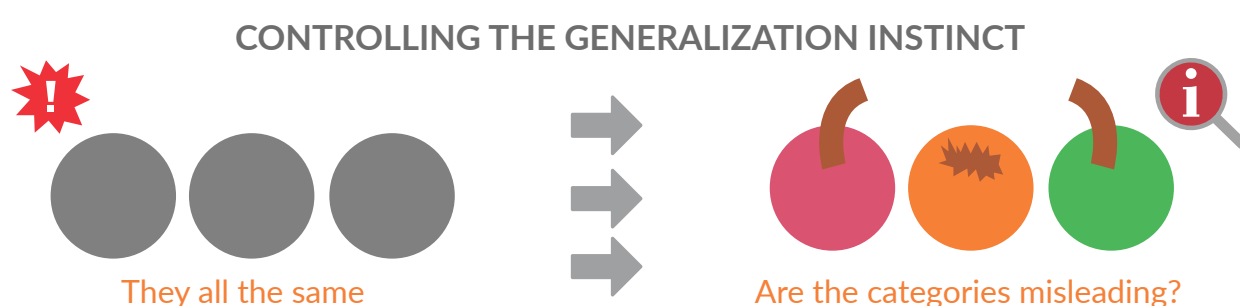
i To control this instinct, be mindful of how you're using categories to explain or predict something.

- Question if your categories are relevant. Look for differences *within* groups, as well as similarities *across* groups. Split large groups into smaller, more meaningful categories (e.g. not everyone in China and Africa have the same income levels). Replace categories that are over-simplified and misleading, like how Rosling replaced “developing countries” and “developed countries” with the 4 income levels.

KEY QUOTES

“Strategic business planners need a fact-based worldview to find their future customers.”

- Beware of extreme examples or vivid images that may be memorable but represent the exception rather than the norm. Be mindful that the word “majority” can mean anything from 51%- 99%.
- Broaden your perspective. When you can't make sense of a solution or practice, consider if it could be smart from a different angle, rather than assume people are stupid. When you travel, stay with people from other income levels rather than in tourist hotels. Examine the photos from Dollar Street to understand how people from different income groups across the world live.



7. The Destiny Instinct



We tend to assume that (a) the destinies of people, cultures, countries etc. are predetermined by certain factors, and (b) such factors are fixed and unchanging. As a result, we assume that other people, groups or countries will always behave in a particular way and are destined to succeed or fail.

! In reality, societies and cultures are constantly shifting.

- Over the last 100 years, societal structures, values, and practices have changed everywhere. For example, the strong support for women's rights in Sweden today is a huge shift from the predominantly patriarchal values of the 1960s. Likewise, life expectancy in several large African countries have improved to be above world average.
- We also shouldn't assume that all Western countries will continue to grow steadily, nor that they will always stay ahead of the “developing countries”.

“Societies and cultures are not like rocks, unchanging and unchangeable. They move.”

KEY QUOTES

 To control this instinct, realize that gradual shifts may not be obvious, but will add up over time to create massive change.

- Slow change $\neq$ no change. A gradual improvement of 1% a year will accumulate to 200% improvement in 70 years.
- Realize that knowledge can go out of date quickly and seek to update your knowledge regularly.
- Look for examples of cultural change which can be found everywhere. For example, speak with your grandparents about their lives in the past and compare their values with yours.

CONTROLLING THE DESTINY INSTINCT



It'll never change



Small change $\neq$ No change



8. The Single Perspective Instinct



We tend to focus on single causes or solutions—such singular perspectives are easier to grasp and make our problems seem easier to solve.

 Unfortunately, singular worldviews are completely misleading, and you shouldn't trust any one information source. For example:

- Experts are just as susceptible to dramatic filters as other people. They may also be biased toward their area of expertise and lack knowledge in other problem areas.
- Activists tend to play up the scale of the problems they are devoted to solving (e.g. human rights, animal protection), and play down the progress that has already been made.
- Idealogues tend to narrowly present their ideology (e.g. equality, free markets) as the best way to solve the world's problems, when there's always more than 1 way to solve any problem.

KEY QUOTES

"Challenge the idea that today's culture must also have been yesterday's, and will also be tomorrow's."

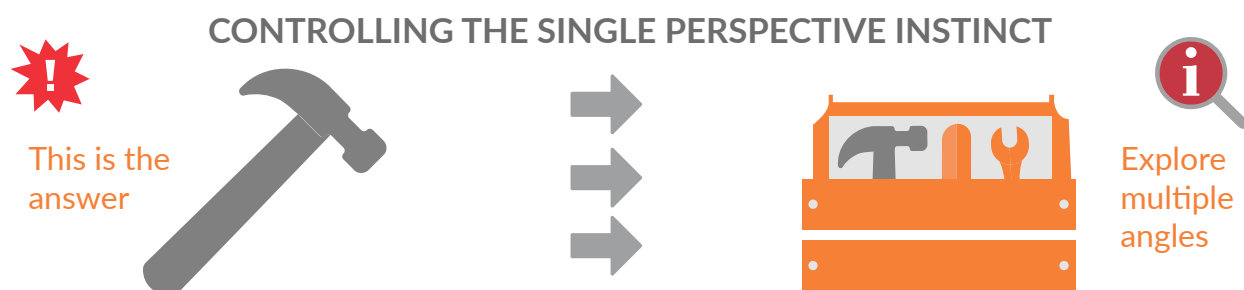
"Great knowledge can interfere with an expert's ability to see what actually works."

"Beware of simple ideas and simple solutions. History is full of visionaries who used simple utopian visions to justify terrible actions."

- Likewise, numbers help us to understand the world, but we can't explain everything using numbers. Medicine is vital in battling diseases, but it's not the only way to save lives. Democracy may be a great way to run a country, but it's not the only way to deliver economic and social progress.

 To control this instinct, examine problems from multiple angles.

- Work with a toolbox, not just a hammer. Don't let your expertise blind you to alternative views, and don't rely on 1 tool to solve all problems. Work with people who're strong in other fields. Use numbers but don't rely only on them.
- Test your ideas or hypotheses on a wide range of samples, including people who disagree with you.



9. The Blame Instinct



When something goes wrong, we instinctively blame it on someone or something. If you slip on a puddle of water, your first instinct may be to blame the person who spilled the water, or the cleaner who failed to mop it up. By focusing on these factors, you miss the simple fact that you slipped because you weren't paying attention to your steps.

 The blame instinct distracts you from the real causes behind the problems and hinders your ability to solve them. When you blame the “bad guys” (e.g. unscrupulous businessmen or dishonest journalists), you:

- Exaggerate the influence of a single individual or group;
- Stop thinking about the problem in depth; and

KEY QUOTES

“No single measure of a good society can drive every other aspect of its development. It's not either/or. It's both and it's case-by-case.”

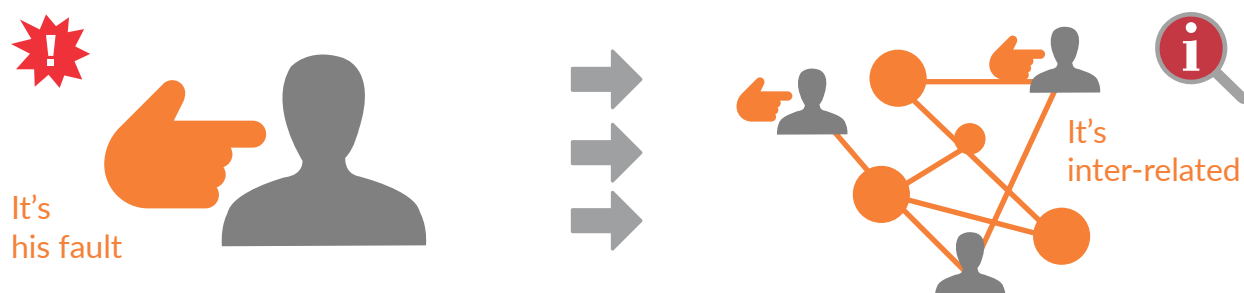
“This instinct to find a guilty party derails our ability to develop a true, fact-based understanding of the world: it steals our focus as we obsess about someone to blame, then blocks our learning because... we stop looking for explanations elsewhere.”

- Neglect the interrelated roles of systems, institutions and technologies. For example, you can launder your clothes with a washing machine not just because of the supplier who sold you the machine, but because of countless people and organizations who led to the invention of the machine, and deliver water and electricity to your home.

 Any issue—especially complex ones like global warming—always involves a system of interrelated causes. It's futile to blame any single entity. Likewise, no single leader or group can claim full credit for a certain outcome.

 To control this instinct, look for causes and systems, instead of villains and heroes. Seek to understand how multiple factors interact in a wider system to create the situation. Ask yourself if the outcome would still have happened without an individual or organization.

CONTROLLING THE BLAME INSTINCT



10. The Urgency Instinct



We tend to rush into a problem or opportunity for fear that there's no time and we may be too late.

 Businesses, political/religious leaders all understand this instinct. That's why we're constantly bombarded by limited-time offers or predictions of impending danger to scare us into action. Unfortunately, when we're stressed or fearful, we cannot analyze the situation properly and tend to make bad, impulsive decisions.

 To control this instinct, realize that what feels urgent usually isn't.

- Calm down and look for relevant and accurate data, so you can validate the claims and understand the real situation.

KEY QUOTES

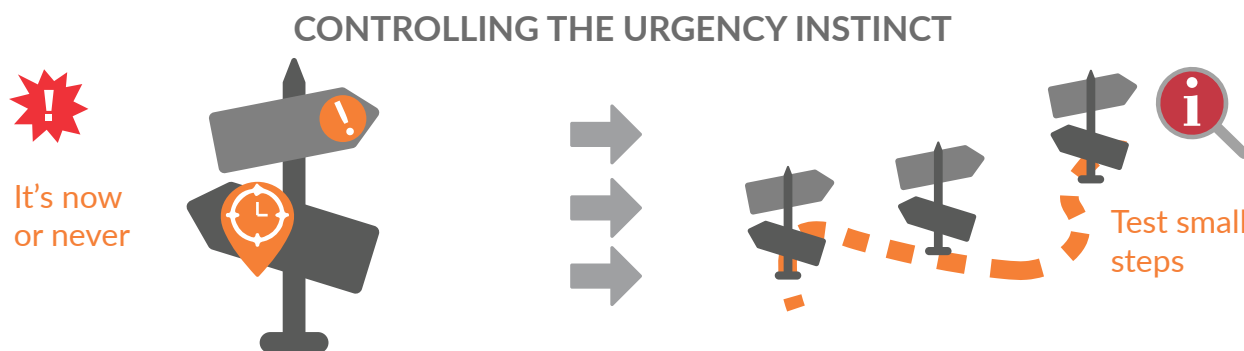
"It's almost always about multiple interacting causes—a system."

"If you really want to change the world you have to understand it. Following your blame instinct isn't going to help."

"When we are afraid and under time pressure and thinking of worst-case scenarios, we tend to make really stupid decisions."

"When you are called to action, sometimes the most useful action you can take is to improve the data."

- Beware of predictions about the future, since most long-term forecasts (be it of the weather, economy or stock markets) are inaccurate.
- Don't jump into drastic solutions, as they may create other unexpected problems. Instead, consider if an idea has been tested, and use step-by-step improvements so you can validate if they're effective.



Using Factfulness in Daily Life

Our dramatic instincts exist for a reason and everyone has them. The key is to manage these instincts, so we can think factfully and find better solutions based on a real understanding of the world.

- Education: Teach our children about the 4 income levels so they can understand the world they live in. Help them to use factfulness as a part of critical thinking, challenge their assumptions and stay open minded to new information. As adults, we should also constantly update our knowledge. Improve how you filter and consume the news, rather than rely on journalists, activists, and politicians to present factual and balanced reports.
- Business: By using facts rather than assumptions, businesses can make better decisions about consumers, recruitment, investments etc. Start by identifying the most important facts your organization should have, and ask how many people actually know them.

KEY QUOTES

"When we have a fact-based worldview, we can see that the world is not as bad as it seems—and we can see what we have to do to keep making it better."

"Ultimately, it is not journalists' role, and it is not the goal of activists or politicians, to present the world as it really is...it is up to us as consumers to learn how to consume the news more factfully."